

# Chapter 61. Nitric Oxide as an Adjunct

Klaus Lewandowski

## Nitric Oxide as an Adjunct: Introduction

As the saying goes, unexpected scientific discoveries are often the most important. The “principle of limited sloppiness,” a term coined to describe fortuitous or accidental discoveries, hit in the 1970s, when Zawadski, a technician in the laboratories of Robert F. Furchgott, failed to follow his superior’s directions correctly and did not remove the endothelium in a rabbit aorta preparation. In this preparation, acetylcholine caused potent relaxation whereas contraction was expected. Shortly thereafter, it was established that acetylcholine was acting on endothelial cell receptors to produce a substance that could diffuse to the vascular smooth muscle and initiate its relaxation.<sup>1</sup> This substance was called endothelium-derived relaxing factor. It took another 8 years for independent working groups to confirm that the chemical structure of endothelium-derived relaxing factor was identical to that of [nitric oxide](#) (NO).<sup>2,3</sup>

The scientific and global community honored the substance itself and its discovery by naming NO “Molecule of the Year” in 1992.<sup>4</sup> The Nobel Prize in Physiology or Medicine for 1998 was awarded jointly to Robert F. Furchgott, Louis J. Ignarro, and Ferid Murad, for their breakthroughs concerning “[nitric oxide](#) as a signaling molecule in the cardiovascular system.”

NO is a colorless and odorless gas. It is a toxic air pollutant, present in motor vehicle exhaust and power plant effluent. The gas is found in the atmosphere in the range of 10 to 500 parts per billion (ppb), and locations with heavy vehicular traffic can exceed 1.5 parts per million (ppm). In the hot cone of a glowing cigarette, concentrations of 1000 ppm were measured in a 40-mL puff. NO is a free radical; it quickly reacts with [oxygen](#) (O<sub>2</sub>) to form poisonous nitrogen dioxide.

In the 1980s it became evident that NO is an essential molecule that regulates a wide range of human physiologic processes. Early studies revealed that NO is produced in endothelial cells and diffuses to vascular smooth muscle cells, where it mediates relaxation. Further studies demonstrated that the substance controls several other physiologic systems, including the immune system, platelet aggregation, and neurotransmission. The focus of this chapter is the prominent role of NO in respiratory physiology and its therapeutic application by inhalation.

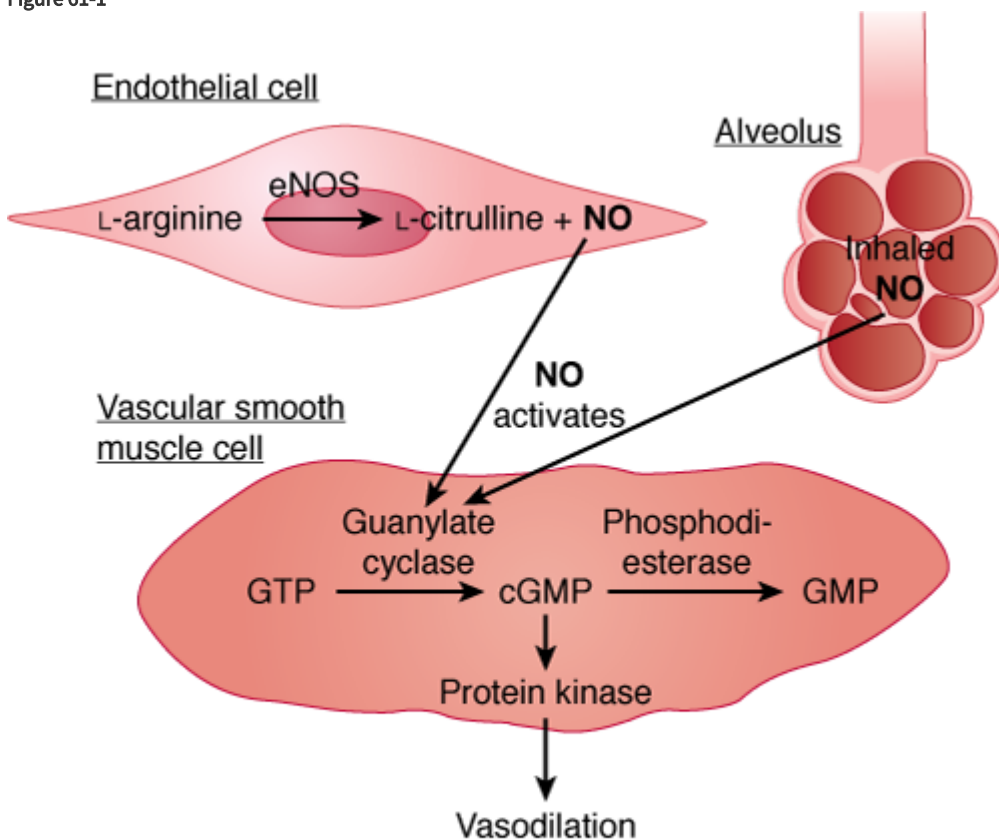
## Endogenous Nitric Oxide in the Respiratory System

## Endogenous Nitric Oxide Synthesis

Endogenous NO is produced by the enzyme system, **nitric oxide synthase (NOS)**. In human subjects, NOS activity can be found in the epithelium of nasal and paranasal mucosa, the bronchial epithelium, type II alveolar epithelial cells, airway nerves, inflammatory cells, airway and vascular smooth muscle cells, and endothelial cells. Three isoforms of the enzyme have been identified: the constitutive neuronal NOS, the inducible NOS (iNOS) that is incited by cytokines, and the constitutive endothelial NOS. There is evidence that a fourth isoform, mitochondrial NOS exists, which has important functions in cellular metabolism.

NO generated by neuronal NOS in the peripheral nervous system acts as a neurotransmitter that modulates smooth muscle relaxation in the respiratory tract. Inflammatory cells express iNOS that enhances NO synthesis. It is supposed that NO plays a self-regulatory role in host-defense and inflammatory processes. Yet, there are conflicting results whether NO mediates proinflammatory or antiinflammatory effects. Concurrently, overproduction of NO may also be associated with the worsening of certain infectious diseases. NO formed by endothelial NOS in vascular endothelial cells regulates pulmonary and systemic vascular tone. Mitochondrial NOS is assumed to provide substantial amounts of cardiac NO responding to heart hypoxia. **Figure 61-1** shows the schematic pathway of NO signal transduction.

Figure 61-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: [www.accessanesthesiology.com](http://www.accessanesthesiology.com)

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Endogenous or inhaled **nitric oxide (NO)** mediates vasodilation of vascular smooth muscle cells. NO is endogenously produced in endothelial cells of the pulmonary vasculature from the amino acid L-arginine, which is converted to L-citrulline and NO by the enzyme **nitric oxide synthase**. NO expressed by the

endothelial cells, or inhaled NO, diffuses rapidly into the vascular smooth muscle cells, where it activates soluble guanylate cyclase, which converts guanosine triphosphate (*GTP*) into cyclic guanosine monophosphate (*cGMP*). The high intracellular concentration of cGMP relaxes the smooth muscle via cGMP-dependent protein kinase. cGMP is inactivated by the enzyme phosphodiesterase, which catalyzes the conversion of cGMP to guanosine monophosphate (*GMP*). *eNOS*, endothelial nitric oxide synthase.

## Endogenous Nitric Oxide Concentrations in the Airways

The concentrations of NO in healthy human airways differ depending on the measurement site. Approximately 100 ppb were measured in the nasopharynx of healthy nonsmoking volunteers during nose breathing. During mouth breathing, even higher concentrations of 650 ppb were seen in the nasopharynx. The highest NO concentrations (1 to 30 ppm) were detected in the paranasal sinuses.<sup>5</sup> In the trachea of intubated patients the NO concentrations were markedly lower; they ranged between 5 and 10 ppb. The cilia of the epithelial cells of the maxillary sinuses contain high amounts of iNOS and can be viewed as a major production site of NO in the respiratory tract.<sup>6</sup>

## Function of Endogenous Nitric Oxide in the Respiratory System

Endogenous NO was suggested to be an important signaling molecule in numerous physiologic processes. Autoinhaled NO from the paranasal sinuses is able to induce selective pulmonary vasodilation in ventilated areas. The blood flow through well-aerated lung areas with higher intraalveolar O<sub>2</sub> concentrations increases and ventilation–perfusion mismatch is antagonized.

Endogenous NO is also involved in host defense and has direct microbicidal effects. Especially the high levels of endogenous NO in the paranasal sinuses may be active in local nasopharyngeal host defense against bacterial or viral invaders. Other findings substantiate that endogenous NO increases airway mucus secretion. Ciliary beat frequency, responsible for microbial clearance, is also enhanced by iNOS stimulators.

Bronchodilation is suppressed by NOS inhibitors, indicating that endogenous NO modulates basal bronchial tone. There is evidence that endogenous NO may protect the airways of asthmatic patients from bronchoconstriction.<sup>7</sup> The high levels of NO, measured in the exhaled air of patients with asthma, however, are believed to mirror the stimulation of iNOS by proinflammatory cytokines and seem to have harmful effects, such as inflammation and increased vascular permeability.

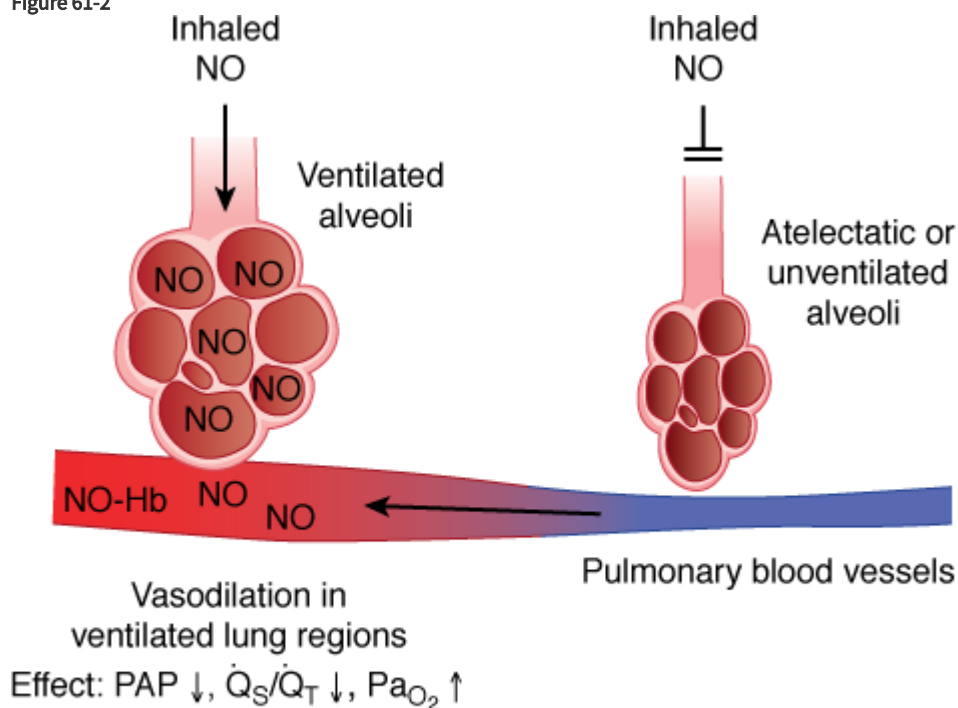
NO is excreted by tumor and host cells and is a factor of promoting angiogenesis and tumor growth. The role of NO in cancer, however, is yet undetermined as it has been attributed both, tumoricidal as well as tumorigenic properties. There are indications that inhibiting iNOS may be beneficial in treatment of certain forms of cancer.

Furthermore, endogenous NO possibly regulates the coagulation system. Formation of endogenous NO also seems to be involved in mediating pulmonary vasodilation during transition of the pulmonary circulation at birth.

## Rationale for the Use of Inhaled Nitric Oxide as an Adjunct

Inhaled **nitric oxide** (iNO) acts as a selective pulmonary vasodilator. As a gas, it reaches only ventilated alveoli and produces relaxation of the accompanying pulmonary blood vessels. iNO acts by producing vasodilation in well-ventilated lung units and redistributing pulmonary blood flow from unventilated to ventilated regions of the lung. This results in improvement of ventilation–perfusion mismatch and oxygenation, in a decrease of pulmonary vascular resistance and pulmonary artery pressure (PAP), as well as in a lower right-ventricular afterload. The activity of iNO is mainly limited to the area of deposition when lower concentrations of iNO are applied because large amounts of the molecule are inactivated by binding to hemoglobin at the moment NO diffuses into the blood vessel. This explains why iNO has almost no systemic side effects and acts predominantly in the pulmonary circulation (Fig. 61-2).

Figure 61-2



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: [www.accessanesthesiology.com](http://www.accessanesthesiology.com)

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Inhaled NO selectively produces vasodilation in ventilated alveoli. iNO produces vasodilation in well-ventilated lung units and redistributes pulmonary blood flow from unventilated to ventilated lung regions. In patients with ventilation–perfusion mismatch (e.g., patients with acute respiratory distress syndrome), this leads to improvement in oxygenation and reduction of intrapulmonary right-to-left shunt. Selective vasodilation in the pulmonary circulation causes significant reduction of elevated pulmonary artery pressure; thus, iNO is a valuable therapy in patients with all kinds of pulmonary hypertension. Systemic vasodilation does not occur to a noteworthy extent because large amounts of NO rapidly bind to hemoglobin and are thereby inactivated. *Hb*, hemoglobin; *iNO*, inhaled **nitric oxide**; *NO*, **nitric oxide**;  $Pa_{O_2}$ , arterial partial pressure of **oxygen**; *PAP*, pulmonary arterial pressure;  $\dot{Q}_s/\dot{Q}_T$ , intrapulmonary right-to-left shunt.

## Indications and Outcome of Inhaled Nitric Oxide Therapy

## Adults and Older Children

Inhaling low concentrations of NO causes rapid and safe reduction of an elevated PAP, and improves the impaired oxygenation in many patients without causing systemic hypotension. In the early 1990s, iNO became an innovative treatment option for patients with acute respiratory failure and pulmonary hypertension.

In patients with acute respiratory distress syndrome (ARDS), improvement of oxygenation and reduction of PAP is an important therapeutic goal. iNO selectively enhances perfusion in ventilated lung areas and counteracts the ventilation–perfusion mismatch typical of this condition. Because NO works only in aerated lung tissue, measures that recruit previously collapsed alveoli can enhance the beneficial effect of NO. The overall effect of iNO is often an impressively increased arterial partial pressure of oxygen ( $PA_{O_2}$ ) and a reduction of the elevated PAP. Several enthusiastic reports encouraged the hope that iNO would prove to be the promising new therapy that would ultimately improve the low survival rates in ARDS. Disappointingly, an improved outcome of ARDS patients could not be observed. A meta-analysis of twelve randomized controlled trials that included 1237 patients revealed that iNO significantly improves oxygenation during the first days of inhalation, although no benefit on survival could be detected. Treatment with iNO was further burdened with an increased risk of renal dysfunction.<sup>8</sup>

The significant improvements in oxygenation and reduction of elevated PAP by iNO, however, could not be denied, and prompted scientists to study the effect of iNO in other diseases associated with severe hypoxia or pulmonary hypertension. [Table 61-1](#) presents a literature survey, focusing on iNO therapy in several diseases or conditions in adults or older children. It further provides expert recommendations for the application of iNO in the respective patient collectives. Synopsis of the information compiled in the table allows for the conclusion, that in most cases of severe oxygenation impairment or high PAP, iNO is effective. When considering iNO as a possible therapeutic option in adult patients, it is important to recognize that certain pathophysiologic variables may significantly improve, although in no instance was survival clearly enhanced.

**Table 61-1: iNO Treatment in Adults and Older Children: Benefit and Expert Recommendations**

| Diagnosis                                                     | Effect of NO Inhalation                                                                                                                                                                                                                                                                                                                                                                    | Expert Recommendations                                                                                                                                                                                                                                     |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heart disease, heart failure (iNO for diagnostic purposes)    | Pulmonary vascular reactivity testing with O <sub>2</sub> and iNO can identify patients with pulmonary hypertension suitable for corrective cardiac surgery or heart transplantation. <sup>52</sup> Patients with severe heart failure may experience pulmonary edema with inhalation of NO, probably because of increased left atrial filling after pulmonary vasodilation. <sup>53</sup> | iNO trial for identification of patients suitable for cardiac surgery or heart/lung transplantation is recommended. iNO testing in patients with left-heart dysfunction is dangerous, heart function should be optimized before iNO testing. <sup>20</sup> |
| Pulmonary arterial hypertension (iNO for diagnostic purposes) | iNO decreases PAP effectively in some patients. The acute decrease of PAP with iNO is the best predictor of long-term response to oral vasodilator treatment. <sup>54</sup> iNO can identify responders for a long-term treatment with calcium-channel blockers. <sup>55</sup>                                                                                                             | iNO is recommended for acute vasodilator testing in a dosage of 10 to 20 ppm. <sup>20</sup> Insufficient data to recommend long-term inhalation of NO. <sup>20</sup>                                                                                       |

| Diagnosis                                    | Effect of NO Inhalation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Expert Recommendations                                                                                                                                                                                |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thromboembolism                              | In four cases of pulmonary embolism, iNO led to improvement of pulmonary hemodynamics and oxygenation. <sup>56</sup> In four further patients with acute massive pulmonary embolism, inhaled NO rapidly improved pulmonary and systemic blood pressures, heart rate, and gas exchange. <sup>57</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | No routine use of iNO in thromboembolic disease recommended because of insufficient data. In patients with severe right-ventricular failure or severe hypoxemia, iNO may be beneficial. <sup>20</sup> |
| Sickle cell disease                          | Most common complications of sickle-cell disease are the vasoocclusive pain crisis and the acute chest syndrome, a form of lung injury, as well as hemolysis. Hemolytic anemia is associated with pulmonary hypertension. <sup>58</sup> Results from two small randomized, placebo-controlled trials suggested that in severe vasoocclusive crisis, inhalation of 80 ppm NO decreased pain scores and morphine use. <sup>59,60</sup>                                                                                                                                                                                                                                                                                                                                        | No routine use of iNO recommended because of insufficient data. <sup>20</sup>                                                                                                                         |
| Chronic obstructive pulmonary disease (COPD) | Effects of iNO in patients with COPD are contradictory. In one study, iNO improved oxygenation and reduced PAP while right-ventricular ejection fraction increased. <sup>61</sup> In another study, iNO reduced PAP but did not improve right-ventricular ejection fraction or arterial oxygenation in patients with acute respiratory failure caused by acute exacerbation. <sup>62</sup> In a third study, however, addition of iNO to inhaled oxygen did not improve or worsen arterial partial oxygen pressure, but caused a significant decrease in mean PAP. Cardiac output increased. Long-term use of iNO was effective. <sup>63</sup> In ten volunteers with very severe COPD iNO in concentrations of 40 to 40,000 ppb did not improve oxygenation. <sup>64</sup> | No evidence of a clinical benefit of iNO in patients with COPD. <sup>20</sup>                                                                                                                         |

| Diagnosis                                                       | Effect of NO Inhalation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Expert Recommendations                                                                                                                                                                                                                               |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| High-altitude pulmonary edema (HAPE)                            | In mountaineers prone to HAPE, iNO produced a marked decrease in PAP. In subjects with radiographic evidence of pulmonary edema, iNO improved oxygenation. <sup>65</sup> In fourteen patients with severe HAPE, iNO reduced PVR by 36% compared with room air. PVR fell by 54% when iNO was combined with 50% oxygen. $Pa_{O_2}$ increased by 14% when iNO was applied. <sup>66</sup> In a rat model of HAPE, iNO improved survival. <sup>67</sup>                                                                                                                                                                                        | No recommendation because of insufficient data.                                                                                                                                                                                                      |
| Myocardial infarction and cardiogenic shock                     | In thirteen patients with right ventricular myocardial infarction and cardiogenic shock iNO significantly decreased PAP and PVR, and increased CI by 24%. <sup>68</sup> Application of iNO before and during coronary reperfusion is able to reduce infarct size (animal experiment). <sup>69</sup>                                                                                                                                                                                                                                                                                                                                       | No recommendation because of insufficient data.                                                                                                                                                                                                      |
| Right-ventricular failure (RVF)                                 | RVF responds favorably to afterload reduction. In RVF after myocardial contusion or after right-ventricular myocardial infarction with cardiogenic shock, iNO rapidly improved hemodynamics. <sup>68,70</sup>                                                                                                                                                                                                                                                                                                                                                                                                                             | No recommendation because of insufficient data.                                                                                                                                                                                                      |
| Perioperative pulmonary hypertension and RVF in cardiac surgery | Cardiopulmonary bypass reduces NO production in pulmonary tissue that leads to pulmonary vasoconstriction and right-ventricular dysfunction postoperatively (animal experiment). <sup>71</sup> Inhalation of NO during and after cardiopulmonary bypass diminishes the release of markers of myocardial injury. Left ventricular dysfunction during and immediately after cardiopulmonary bypass is antagonized. <sup>72</sup> In twenty-three patients with pulmonary hypertension, treated with iNO immediately after cardiopulmonary bypass, iNO significantly reduced PAP and PVR and lead to increased cardiac output. <sup>73</sup> | In patients with perioperative acute right-ventricular dysfunction and pulmonary hypertension, RVF should be optimized first by conventional measures before a trial of iNO should be undertaken. iNO doses >20 ppm have no advantage. <sup>20</sup> |



| Diagnosis                               | Effect of NO Inhalation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Expert Recommendations                                                                                                                                                                                                                                                                                      |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Left-ventricular assist devices (LVADs) | <p>Pulmonary hypertension is frequent in patients treated with LVADs and may lead to right-ventricular dysfunction. Eleven patients with LVADs and pulmonary hypertension were randomized to iNO (20 ppm) and control therapy. iNO precipitated significant reductions in PAP and increased LVAD flow. In control patients, no hemodynamic improvement was recorded.<sup>74</sup> A recent randomized controlled trial studied the effect of iNO in forty-seven patients with implanted LVADs. Only in the iNO group did PVR decrease significantly, from <math>311 \pm 35</math> to <math>225 \pm 17</math> dyn sec cm<sup>-5</sup> (<math>p &lt; 0.01</math>).<sup>75</sup></p>                                                                                                       | <p>Expert panel believes that iNO improves pulmonary hemodynamics in patients with inadequate left-sided flow during use of LVADs and pulmonary hypertension refractory to conventional maneuvers. In this situation, they recommend application of iNO among other vasodilator therapies.<sup>20</sup></p> |
| Heart transplantation                   | <p>Pulmonary hypertension is a frequent problem during heart transplantation and may contribute to life-threatening right-heart failure. In heart transplant recipients with pulmonary hypertension, iNO given in the postoperative period selectively reduced PVR and enhanced right-ventricular stroke work. Compared with a historical cohort, the NO treated group had better survival rates.<sup>76</sup> Fourteen patients with either heart transplantation or lung transplantation received iNO in the operating room when pulmonary hypertension, refractory hypoxemia, or right-ventricular dysfunction were present. Inhalation of 20 ppm NO lowered PAP and central venous pressure, increased cardiac index, and improved mixed venous oxygen saturation.<sup>77</sup></p> | <p>iNO is used by several institutions with experience in cardiac transplantation. They recommend it as standard therapy for heart transplantation complicated by an elevated PVR.<sup>20</sup></p>                                                                                                         |

| Diagnosis                  | Effect of NO Inhalation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Expert Recommendations                                                                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| One-lung ventilation (OLV) | Hypoxia is a frequent complication during OLV. Sixteen patients who developed hypoxemia during OLV were randomized to iNO (20 ppm) or control groups (nitrogen). iNO when administered to the dependent lung was not superior to nitrogen. <sup>78</sup> In patients with pulmonary hypertension during OLV, iNO caused a significant reduction of mean PAP and improved oxygenation in patients with severe hypoxemia. <sup>79</sup> A combination of iNO with almitrine significantly improved oxygenation during OLV. <sup>26</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Expert panel does not recommend routine use of iNO during OLV. Only in case of severe hypoxemia, refractory to conventional therapy, iNO may be beneficial. <sup>20</sup> |
| Major lung resection       | Postpneumonectomy pulmonary edema is a severe complication of lung resection and afflicted with high mortality rates. In a patient with major lung resection iNO was successfully applied to treat postpneumonectomy pulmonary edema (case report). <sup>80</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | No recommendation because of lack of data.                                                                                                                                |
| Lung transplantation       | There remains controversy whether iNO can prevent ischemia–reperfusion injury in lung-transplant recipients. Some studies performed in the 1990s reported that iNO reduced reperfusion injury in transplanted lungs. Another study, however, showed that prophylactic iNO does not prevent reperfusion injury. During reperfusion, however, patients with reperfusion injury experienced improved oxygenation and reduction of PAP. <sup>81</sup> Contrary to earlier studies, a large randomized, controlled trial could also not detect a significant effect on physiologic variables or outcomes in the group of iNO-treated patients, in whom iNO was initiated 10 minutes after reperfusion. <sup>82</sup> The occurrence of acute graft rejection, however, was less frequent in the iNO group in comparison with historical controls. <sup>83</sup> Recent randomized clinical trials could not detect a benefit of a prophylactic administration of iNO for prevention of primary graft failure, however, in case of development of a hypoxic phase during primary graft failure, iNO may reduce the need for extracorporeal membrane oxygenation in lung transplant patients. <sup>84</sup> | No evidence that iNO prevents reperfusion injury after lung transplantation. <sup>20</sup>                                                                                |

| Diagnosis                                                           | Effect of NO Inhalation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Expert Recommendations                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acute lung injury (ALI)/ acute respiratory distress syndrome (ARDS) | iNO was found to reduce elevated PAP, improve oxygenation, and decrease intrapulmonary right-to-left shunt during the first days of treatment. <sup>31</sup> A meta-analysis of twelve randomized controlled trials that included 1237 patients revealed that iNO improves oxygenation during the first days of inhalation. No benefit on survival could be detected. Risk of renal dysfunction was increased in the iNO patients. <sup>8</sup> These findings were confirmed by a large systematic review of the Cochrane Collaboration. <sup>29</sup> | No routine use of iNO in ALI/ARDS. Trial of iNO as a rescue treatment in case of life-threatening hypoxemia. <sup>20</sup>                                                                                                                                |
| Influenza                                                           | Patients with H1N1 influenza very rapidly develop respiratory failure. Thirty-two patients of the H1N1 influenza epidemic 2009 in Spain were evaluated and 25% received iNO as a rescue therapy. <sup>85</sup> In Canada, 168 critically ill patients with influenza A (H1N1) were admitted to thirty-eight intensive care units and prospectively evaluated; 13.7% of the patients received iNO as a rescue therapy. <sup>86</sup>                                                                                                                     | ICUs treating influenza patients should provide advanced ventilatory support and rescue therapies including iNO. <sup>87</sup>                                                                                                                            |
| Inhalation injury                                                   | There are only few reports on use of iNO in burn patients with inhalation injury. <sup>88-90</sup> All studies reported an improved oxygenation with iNO therapy. An immediate and stronger early response to NO inhalation may eventually predict recovery, however, all studies were only performed in a small series of patients with the respective weakness of the reported results.                                                                                                                                                               | No routine use of iNO in inhalation injury. Trial of iNO as a rescue treatment in case of life-threatening hypoxemia after conventional treatment has been optimized. In case of early response to iNO, the inhalation should be continued. <sup>91</sup> |

| Diagnosis | Effect of NO Inhalation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Expert Recommendations                                                                |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Asthma    | Inhalation of NO results only in a minor relaxation of airway tone in adults. <sup>92,93</sup> In children with asthma, iNO has no apparent bronchodilatory effect. <sup>94</sup> In case of severe status asthmaticus, however, iNO might have a bronchodilatory effect (case report). <sup>95</sup> In five children with life-threatening status asthmaticus who required MV and did not respond to maximal medical management, iNO decreased $\text{Pa}_{\text{CO}_2}$ significantly. Four children survived. <sup>96</sup> | No recommendation as to date no controlled studies on this topic have been performed. |

*Abbreviations:* CI, cardiac index; ICU, intensive care unit; iNO, inhaled nitric oxide; PAP, pulmonary artery pressure; PVR, pulmonary vascular resistance.

What are still suitable indications for iNO in adults or older children? It may be indicated in patients who are in a phase of severely impaired gas exchange that is unresponsive to maximal medical therapy. In such settings, application of iNO can significantly enhance pulmonary gas exchange and thereby prevent hypoxic organ damage. Moreover, iNO can be tried in all patients suffering from severe pulmonary hypertension. In most cases, elevated PAP drops and relieves the right heart. When considering possible iNO treatment, however, it is strongly recommended to optimize conventional treatment before a trial of iNO is undertaken.

## Infants

Application of iNO yielded the best results in critically ill infants. In 1999, the use of iNO was approved by the United States Food and Drug Administration (FDA) for the treatment of term and near-term newborns (>34 weeks of gestation) with hypoxic respiratory failure associated with pulmonary hypertension.

In neonates with respiratory failure, persistent pulmonary hypertension of the newborn (PPHN) is common. PPHN is characterized by elevated pulmonary resistance, pulmonary vasoconstriction, and altered vascular reactivity. Desaturated blood circulates partly through an extrapulmonary right-to-left shunt, across the foramen ovale and ductus arteriosus. Since 1970, extracorporeal membrane oxygenation (ECMO) has been the treatment of choice if PPHN is present. Survival rates of up to 80%<sup>9</sup> were achieved with ECMO therapy whereas survival with conservative therapy was approximately 50%.<sup>10</sup> In hypoxemic newborns with PPHN, clinical studies indicate that iNO increases systemic  $\text{O}_2$  levels, decreases PAP, and mitigates ventilation-perfusion mismatch. Randomized, placebo-controlled trials of iNO in neonates with PPHN failed to show a significant decrease in mortality rates in the iNO group; iNO therapy did, however, reduce the requirement for ECMO.<sup>11,12</sup>

Some years ago, a large, randomized, controlled trial documented the effectiveness of iNO in 207 preterm neonates with respiratory failure. iNO decreased the incidence of chronic lung disease and death.<sup>13</sup> These encouraging findings, however, were not supported by further studies. A recent systematic review of fourteen randomized controlled trials that studied iNO therapy in preterm neonates unveiled a disappointing picture. iNO was not able to significantly decrease mortality rates and early rescue treatment is probably associated with a higher risk of intraventricular hemorrhage.<sup>14</sup> In a selected patient collective of premature infants with a birth weight between 1000 and 1250 g, however, early inhalation of NO might reduce the incidence of bronchopulmonary dysplasia.<sup>15</sup>

Perioperative pulmonary hypertension in infants with congenital heart defects is harmful and compromises their chance of survival. iNO, however, has not proven to reduce mortality or the number of pulmonary hypertensive crises.<sup>16</sup> A trial with iNO could only be recommended in infants with severe pulmonary hypertension in the perioperative setting of congenital heart surgery.<sup>17,18</sup>

From a pathophysiologic point of view, iNO therapy may be harmful for newborns with congenital heart disease dependent on right-to-left shunts and its routine use is not recommended in this setting.<sup>11</sup> Newborns with congestive heart failure and lethal congenital anomalies should also be excluded from iNO therapy.<sup>11</sup>

Table 61-2 presents an overview of the literature on the effect of iNO therapy in several diseases or conditions in infants together with expert recommendations. In a nutshell, physicians caring for infants with hypoxic respiratory failure and PPHN should be familiar with the use of iNO. In newborns it should be used liberally if these indications are present, because ECMO therapy can be avoided in many cases. In preterm infants, however, routine use of iNO is not recommended.<sup>17</sup>

Table 61-2: iNO Treatment in Infants: Benefit and Expert Recommendations

| Application of iNO/Diagnosis                              | Effect of NO Inhalation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Expert Recommendations                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assessment of the reversibility of pulmonary hypertension | When evaluating a patient for corrective heart surgery or heart transplantation, an iNO trial may be useful in determining operability. One hundred twenty-four patients with heart disease complicated by pulmonary hypertension received 100% oxygen and iNO to test pulmonary vascular reactivity. Test results could identify proper candidates for corrective surgery. <sup>52</sup> A recent retrospective trial studied children with pulmonary hypertension who underwent vasodilator testing before heart transplantation. Children who did not respond had increased rates of right-ventricular failure (RVF) and death from RVF. Preemptive use of iNO might decrease frequency of RVF. <sup>97</sup> | Pulmonary reactivity testing with iNO and oxygen is recommended in infants with heart disease complicated by pulmonary hypertension to identify proper candidates for corrective surgery. <sup>18</sup>                                                                  |
| Hypoxia in term and near-term neonates                    | In 1999, use of iNO was approved by the FDA for treatment of hypoxic respiratory failure and pulmonary hypertension in term and near term neonates. The Cochrane Collaboration performed an analysis of fourteen randomized, controlled studies in term and near-term newborns with hypoxia. Use of iNO reduced the need for extracorporeal membrane oxygenation (ECMO) and improved oxygenation. Reduction of mortality was not evident. There were, however, indications that outcome of infants with diaphragmatic hernia was worsened. <sup>12</sup>                                                                                                                                                         | A trial of iNO is favored in newborns $\geq 34$ weeks gestation, $< 14$ days of age and with $\text{Pa}_{\text{O}_2}/\text{Fi}_{\text{O}_2} < 100$ mm Hg and/or an oxygenation index $> 25$ . iNO therapy should begin early in the course of the disease. <sup>11</sup> |

| Application of iNO/Diagnosis                                                                                                                                     | Effect of NO Inhalation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Expert Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hypoxia in preterm neonates                                                                                                                                      | <p>Despite some encouraging early reports, iNO has not proven to be beneficial in preterm neonates. A recent systematic review performed by the Cochrane Collaboration analyzed fourteen randomized, controlled trials investigating iNO therapy in preterm neonates. Rescue therapy with iNO in case of severe hypoxia did not reduce mortality. Routine application of iNO in infants with pulmonary disease also could not reduce mortality. Although early rescue treatment was associated with a 20% higher incidence of intraventricular hemorrhage, this finding was not significant.<sup>14</sup></p> | <p>Routine use of iNO in preterm infants (&lt;34 weeks gestation) is not recommended.<sup>17</sup></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Persistent pulmonary hypertension of the newborn (PPHN) caused by lung parenchyma diseases leading to abnormally constricted pulmonary vasculature <sup>98</sup> | <p>Meconium aspiration syndrome, Respiratory distress syndrome, Pneumonia</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>In 1999, use of iNO was approved by the FDA for treatment of hypoxic respiratory failure and pulmonary hypertension in term and near-term neonates. A recent systematic review for the development of evidence based clinical practice guidelines for use of iNO in newborns, presenting with hypoxemic respiratory failure associated with pulmonary hypertension, revealed the following results<sup>11</sup>: analysis of nine studies that reported mortality detected no significant effect of iNO on this parameter; eight studies evaluated long-term outcomes. Neurodevelopmental outcomes were similar in neonates treated with iNO and controls. Pulmonary outcomes in both groups were also similar. Eight studies reported requirement for ECMO. Meta-analysis of these data revealed less need of ECMO in the iNO groups.</p> |
|                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>A trial of iNO is recommended in newborns <math>\geq 34</math> weeks gestation, &lt;14 days of age and with <math>\text{Pa}_{\text{O}_2}/\text{Fi}_{\text{O}_2} &lt; 100</math> mm Hg and/or an oxygenation index &gt;25. iNO therapy should begin early in the course of the disease.<sup>11</sup> iNO therapy should start with 20 ppm. If there is no response to a 30- to 60-minute trial of iNO, iNO therapy should be terminated. In parenchymal lung disease optimal recruitment of lung tissue should be achieved before</p>                                                                                                                                                                                                                                                                                                       |

| Application of iNO/Diagnosis                                                                     | Effect of NO Inhalation               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | NO inhalation.<br><b>Expert</b><br>Lowest effective<br><b>Recommendations</b><br>dose of iNO and                                              |
|--------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| PPHN due to abnormally remodeled pulmonary vasculature with normal lung parenchyma <sup>98</sup> | Idiopathic PPHN                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | O <sub>2</sub> should be applied. <sup>11</sup>                                                                                               |
| PPHN caused by hypoplastic pulmonary vasculature <sup>98</sup>                                   | Congenital diaphragmatic hernia (CDH) | <p>Patients with CDH seem to gain almost no benefit from iNO treatment. A randomized study in fifty-three term and near-term infants with CDH reported no significant improvement in oxygenation in the iNO group. Although patients of the iNO group needed more ECMO treatment and had a higher death rate than control patients, these findings were not significant.<sup>99</sup></p> <p>Controversial results were offered by further studies. Early surgery and iNO improves the outcome and reduces the requirement for ECMO in the treatment of antenatally diagnosed congenital diaphragmatic hernia.<sup>100</sup> Retrospective data from sixty-one newborns with CDH give evidence that survival rates of infants with CDH have improved over the last 10 years, which was attributed by the study team to new therapies, including iNO.<sup>101</sup> This result is not concordant with data from a similar study in twenty-seven Norwegian neonates with CDH. The patient collective treated with novel therapeutic measures including iNO had a higher mortality than that treated before 1997, when these therapy options were not available.<sup>102</sup></p> | No convincing benefit for newborns with congenital diaphragmatic hernia. Routine use of iNO in these patients is not indicated. <sup>11</sup> |



| Application of<br>iNO/Diagnosis     | Effect of NO Inhalation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Expert<br>Recommendations                    |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Bronchopulmonary<br>dysplasia (BPD) | <p>BPD is a chronic lung disease mainly of preterm infants, that often develops after postnatal injury by therapeutic measures, such as mechanical ventilation and high inspiratory oxygen concentrations.<sup>98</sup> Pulmonary hypertension and reduction of alveolar–capillary surface area are significant contributors to high mortality in these patients. There are no convincing results that support prophylactic inhalation of NO for prevention of BPD in preterm neonates. Only one study described a markedly lower incidence (30% vs. 60%) of BPD with iNO therapy in premature infants with a birth weight between 1000 and 1250 g.<sup>15</sup></p> | No recommendation because of a lack of data. |

| Application of iNO/Diagnosis                                             | Effect of NO Inhalation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Expert Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Perioperative pulmonary hypertension in congenital heart patients</p> | <p>Pulmonary hypertensive crises postoperatively after congenital heart surgery are dangerous and afflicted with high mortality. One hundred twenty-four infants with congenital heart defects were randomized to iNO or placebo therapy from heart surgery and postoperatively until extubation as a prophylaxis for prevention of pulmonary hypertension. Frequency of pulmonary hypertensive crises was less in infants receiving iNO (RR = 0.66).<sup>103</sup> These findings differ from that of a Cochrane review of four randomized trials that analyzed the effects of postoperative iNO versus placebo. No differences in several outcome variables (i.e., mortality, number of pulmonary hypertensive crises, oxygenation, etc.) between the two groups could be detected.<sup>16</sup></p> | <p>No routine use of iNO in the postoperative management of hypoxic term or near-term infants with congenital heart disease.<sup>11</sup> More data from clinical trials are necessary to recommend routine prophylactic iNO administration in children who are at risk for pulmonary hypertension after congenital heart surgery.<sup>17</sup> A trial of iNO, however, is proposed in patients with significant pulmonary hypertension in the perioperative setting.<sup>17,18</sup></p> |

| Application of iNO/Diagnosis                       | Effect of NO Inhalation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Expert Recommendations                                                                                                                                      |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cavopulmonary circulations, Fontan-type procedures | iNO may be of use in the setting of cavopulmonary circulations. iNO administered in thirteen patients postoperatively after total cavopulmonary connection significantly decreased transpulmonary pressure gradient and improved oxygenation. <sup>104</sup> These findings were confirmed by a prospective randomized trial in forty-six patients with high PVR and impaired oxygenation after Fontan-type procedure. iNO significantly decreased the transpulmonary pressure gradient and improved the oxygenation, however, the combination of iNO with milrinone was even more effective. <sup>105</sup> | After cavopulmonary procedures a trial of iNO could be considered when a high transpulmonary gradient is present in the postoperative period. <sup>18</sup> |

Abbreviations: iNO, inhaled nitric oxide; PVR, pulmonary vascular resistance; RR, relative risk.

## Technique of Nitric Oxide Inhalation

It is strongly recommended to use a delivery device for iNO that is approved for clinical use.<sup>11</sup> The FDA has published a guidance document for approval of NO delivery systems that specifies the technical requirements for safe administration of iNO.<sup>19</sup> Important specifications are that the gas be supplied in a constant concentration during inspiratory flow; administered iNO concentrations should not vary with flow fluctuation within the inspiratory phase of the respiratory cycle; for regulation of iNO concentrations, a range of 0 to 80 ppm should be allowed; the system should be specified for connection with pharmaceutical grade NO gas cylinders containing 400 to 800 ppm NO in nitrogen; contact time of NO and O<sub>2</sub> should be short to limit NO<sub>2</sub> formation; the system should further be equipped with NO, NO<sub>2</sub>, and O<sub>2</sub> gas analyzers and alarm functions to warn the intensivist against too low or too high levels of NO and O<sub>2</sub> concentrations, discontinuation of gas flow, or accumulation of NO<sub>2</sub>.<sup>11,19</sup> NO analyzers should be able to measure iNO concentrations as low as 1 ppm;<sup>19</sup> and a manual back-up NO supply device for emergency or transport reasons should be available near the ventilator to ensure continuous delivery of the therapeutic gas.<sup>20</sup>

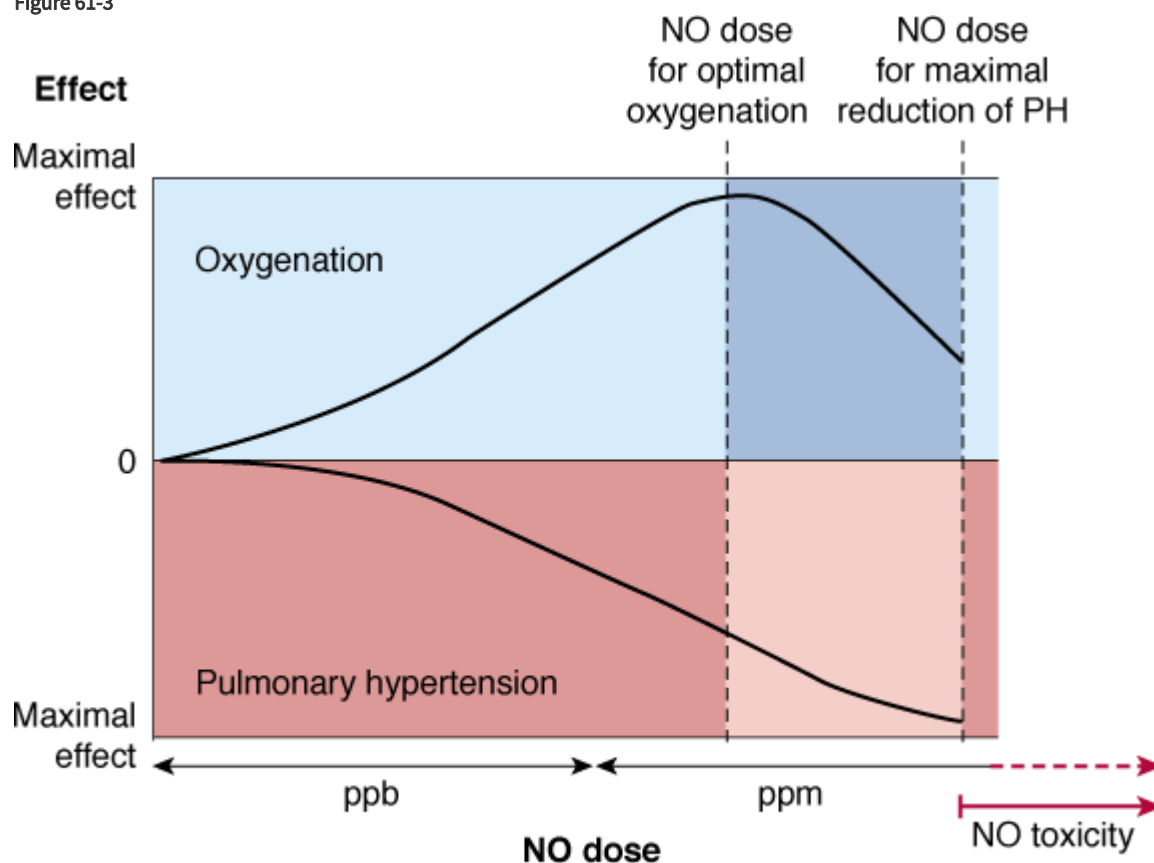
In clinical practice, iNO delivery devices are generally connected to anesthesia and intensive care ventilators, to high-frequency ventilators, to transport ventilators, and manual ventilation systems. In spontaneously breathing, nonintubated infants, iNO therapy is also established by nasal cannula and continuous positive

airway pressure systems. It is recommended to check the manufacturers' specifications before connecting the iNO system to any ventilation device.<sup>11</sup>

## Dosage of Inhaled Nitric Oxide

Dose–response studies in patients with ARDS show that very low concentrations of iNO, only 10 ppb, are able to enhance oxygenation; higher concentrations are necessary to decrease the elevated PAP. The respective effective doses whereby 50% of individuals will show the expected response to iNO are 100 ppb and 2 to 3 ppm. Improvement in oxygenation reached a maximum at 10 ppm and deteriorated at higher NO doses; PAP further decreased with higher NO concentrations. Studies were stopped at 100 ppm.<sup>21</sup> Data suggest that higher doses of NO reach blood vessels supplying partially or nonventilated lung areas. For exact dosage, the registration of an individual dose–response curve for each patient is indispensable (Fig. 61-3). This approach is also desirable in patients with indications other than ARDS.

Figure 61-3



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Schematic dose–response curve for patients with ARDS treated with iNO. Low doses of iNO (in the parts per billion range) are able to improve oxygenation whereas higher levels of iNO are necessary to significantly decrease the elevated PAP. Oxygenation improves to a maximum and then deteriorates at higher NO concentrations. Maximal reduction of PAP needs even higher doses of NO. It is recommended to verify the dose–response characteristics for each individual patient starting before NO treatment. iNO, inhaled **nitric**

oxide; *NO*, nitric oxide; *PAP*, pulmonary arterial pressure; *PH*, pulmonary hypertension; *ppb*, parts per billion; *ppm*, parts per million.

For improvement of oxygenation in adult patients with ARDS, iNO concentrations ranging from 1 to 20 ppm are effective, for reduction of mean PAP a dose in the range of 1 to 40 ppm is required.<sup>22</sup> Concentrations of 80 to 100 ppm should not be exceeded.

For neonates, it is recommended to start with a dose of 20 ppm iNO. If an improvement with iNO therapy is observed, the iNO concentration should be reduced to the lowest dose that sustains the response.<sup>11</sup>

## Responders and Nonresponders

Regarding oxygenation and PAP, about a third of critically ill patients are nonresponders and do not react favorably to NO.<sup>22</sup> Reasons include persistent pulmonary blood flow to dorsal edematous lung areas; physical barriers, such as the presence of exudate or fibrosis; molecular mechanisms, such as increased activity of phosphodiesterase; unfavorable ventilator settings; and, most likely, a number of currently unknown factors. It appears sensible to evaluate a patient's response to iNO on a daily basis because initial nonresponders may convert to responders over time, and vice versa.

## Adjunctive Therapy

When the clinician faces a patient in acute respiratory failure who does not respond to iNO at all or the patient's response is very weak, adjunctive measures can positively change the situation.

Nonresponse to iNO may be related to lack of aerated lung tissue. Measures ensuring that the lungs inflate adequately and therapeutic options maintaining a sufficient end-expiratory lung volume, applied before NO inhalation, may lead to a satisfying NO response.

Re-evaluation of the level of positive end-expiratory pressure represents one of these measures. Recruiting additional alveoli may increase the number of alveoli, whereby NO can easily flow in and exert its beneficial effects. This idea has been tested in experimental and clinical trials. It was shown that when positive end-expiratory pressure led to alveolar recruitment, a rise in  $PA_{O_2}$  of more than 100 mm Hg could be achieved.<sup>23,24</sup>

Ventilating critically ill patients in the prone position may dramatically enhance oxygenation by antagonizing ventilation-perfusion mismatch. Exploiting this mechanism in patients receiving iNO therapy can further enhance oxygenation as compared with iNO in the supine position.<sup>25</sup>

Almitrine bismesylate acts as a pulmonary vasoconstrictor mainly in pulmonary vessels supplying hypoxic lung regions and reduces their perfusion. Theoretically, it was hoped that NO, a selective pulmonary vasodilator, and almitrine, a selective pulmonary vasoconstrictor, would act additively on oxygenation. iNO

combined with almitrine was studied in patients with ARDS and during one-lung ventilation.<sup>26,27</sup> Addition of almitrine to iNO may lead to a further improvement in oxygenation, although a rise of mean PAP of about 3 to 8 mm Hg has to be accepted. Given the potentially harmful effects of this strategy, its routine use is not advisable.

## Side Effects and Toxicology

An adage attributed to the physician, astrologer, and theologian, Philippus Theophrastus Bombastus of Hohenheim, better known as Paracelsus (1493–1541), reads: “All substances are poisons; there is none which is not. The right dose differentiates a poison from a remedy.” Regarding NO, this is surely true. Only a narrow margin exists between an effective and a toxic dose. Direct NO toxicity at clinically relevant doses, however, is uncommon, whereas exposure to NO<sub>2</sub> is potentially toxic.

### Formation of Nitrogen Dioxide

NO reacts with O<sub>2</sub> and forms toxic nitrogen dioxide (NO<sub>2</sub>). High O<sub>2</sub> concentrations combined with high NO concentrations react with the formation of that toxic by-product. Inhaled NO<sub>2</sub> at 2 ppm already affects alveolar permeability and NO<sub>2</sub> concentrations above 10 ppm may induce pulmonary edema, alveolar hemorrhage, methemoglobinemia, hypoxemia, changes in surfactant activity, hyperplasia of type II alveolar cells, intrapulmonary accumulation of fibrin, neutrophils, and macrophages, and eventually death.<sup>28</sup>

Synthesis of harmful concentrations of NO<sub>2</sub> is uncommon, especially when therapeutic doses of NO are inhaled. A study presented by the Cochrane Collaboration that analyzed fourteen randomized trials on acute lung injury/ARDS found no elevated risk for NO<sub>2</sub> formation associated with iNO therapy.<sup>29</sup> In clinical practice, it is recommended to limit the iNO concentration to 10 ppm or less when long-term application is necessary, to reduce possible genesis of toxic NO<sub>2</sub>.<sup>20</sup> Alarm levels should be set to 2 ppm NO<sub>2</sub>.<sup>11</sup>

### Rebound after Nitric Oxide Withdrawal

During weaning from NO, its rapid withdrawal can cause marked rebound pulmonary vasospasm,<sup>30</sup> high intrapulmonary right-to-left shunt, and decreased PAO<sub>2</sub>.<sup>31</sup> Twenty-six percent of patients receiving iNO for acute hypoxemic respiratory failure (responders) exhibited only minimal changes in PAO<sub>2</sub> or hemodynamics after abrupt discontinuation of iNO, whereas 48% exhibited worsening of gas exchange. Another 26% of the patients reacted primarily with hemodynamic side effects. Reapplication of iNO immediately reversed the hemodynamic and oxygenation deterioration.<sup>32</sup> This phenomenon can be explained by downregulation of NO synthase activity in the presence of exogenous NO. In pediatric patients after cardiac surgery, longer continuation of iNO and its lower final concentration are factors that contribute to its successful weaning. Consequently, it seems prudent to slowly wean a patient from iNO. Rebound phenomena may be mitigated

when iNO is not discontinued until a considerable clinical improvement is accomplished and the iNO dose is progressively reduced to 1 ppm before the gas is switched off. It is further recommended to increase  $\text{FI}_{\text{O}_2}$  before iNO cessation.<sup>11</sup>

Alternatively, the clinician can exchange NO for a different pulmonary vasodilator, or wait until the patient's endogenous production recommences. The drug sildenafil, given before discontinuation of iNO, has shown to effectively prevent rebound pulmonary hypertension in children.<sup>33,34</sup>

## Systemic Vasodilation

For many years, a key characteristic of NO seemed to be its selectivity as a pulmonary vasodilator. This characteristic was explained by its rapid binding to the heme moiety of hemoglobin and its very short duration of action. Indeed, in many clinical trials and animal experiments, substantial effects on arterial blood pressure or other systemic circulation variables have not been observed. In some instances, however, reductions in systemic blood pressure and systemic vascular resistance were documented.<sup>35,36</sup> A new insight helps explain this phenomenon: NO can bind reversibly with albumin and hemoglobin and thereby enter the systemic circulation.<sup>37–39</sup> Circulating erythrocytes may be the vehicles that distribute NO to peripheral vascular beds. In areas of low  $\text{O}_2$  tension, erythrocytes offload  $\text{O}_2$  and NO, thereby inducing hypoxic vasodilation.<sup>40</sup> Furthermore, nitrite and nitrate, the oxidative end products of NO metabolism, exert systemic NO effects because they can be recycled to NO in blood and tissues.<sup>41</sup>

## Methemoglobinemia

iNO can combine with hemoglobin to form nitrosyl hemoglobin, which is rapidly oxidized to methemoglobin. High iNO doses are associated with methemoglobin formation. In numerous clinical studies, however, severe methemoglobinemia was not observed with inhalation of NO. Two recent systematic reviews on the use of iNO in acute lung injury, revealed that methemoglobinemia is not a major concern when usual concentrations of iNO are applied.<sup>8,29</sup> It is recommended to measure methemoglobin level 4 hours after initiation of iNO and then once daily.<sup>20</sup> iNO is contraindicated in patients with methemoglobin reductase deficiency.<sup>20</sup> The patient, especially the neonate, should be weaned from iNO if methemoglobin level exceeds 5%.<sup>11</sup>

## Interference with Surfactant

Analyses of surfactants recovered from experimental animals that had received 80 to 120 ppm iNO revealed reduced quality, in terms of capacity to lower surface tension. It is unclear whether this is a consequence of the toxic actions of various NO by-products, such as peroxynitrite and  $\text{NO}_2$ , or if it is mediated by hemoglobin found in the alveoli when there is high-permeability lung edema.<sup>42,43</sup> Further experiments in preterm rabbits revealed that hyperoxia without iNO reduced large surfactant aggregates and surface activity.

Addition of low-dose iNO (14 ppm per nose cannula) prevented or mitigated these deleterious effects on surfactant.<sup>44</sup>

## Effects on Coagulation

Inhibition of human platelet aggregation by iNO has been reported in several studies. In addition, prolongation of bleeding time and inhibition of P-selectin expression with decreased binding of fibrinogen to the platelet glycoprotein IIb/IIIa receptor was demonstrated. Results from a randomized experimental animal study in twelve healthy piglets, however, do not support the hypothesis of an increased risk of bleeding associated with iNO. All coagulation variables including bleeding time were unaffected by iNO.<sup>45</sup> In healthy human volunteers, NO may<sup>46</sup> or may not<sup>47</sup> prolong bleeding time and exert effects on various coagulation variables. In patients with ARDS, iNO did not increase the risk of bleeding.<sup>29</sup> In infants with acute hypoxic respiratory failure, most randomized controlled trials did not report an increase in bleeding disorders with iNO therapy.<sup>11</sup> In premature infants with respiratory distress syndrome, however, a risk of intracranial hemorrhage may be considered.<sup>14</sup>

## Nephrotoxicity

In the kidney, NO controls glomerular and tubular function, the tubuloglomerular feedback, renin release, and the extracellular fluid volume. Large amounts of NO may contribute to progression of renal disease, although NO may have protective qualities in acute renal failure.<sup>48</sup> Two large meta-analyses evaluated the adverse effects of iNO therapy.<sup>8,29</sup> Up to fourteen randomized, controlled trials of iNO treatment in children and adults with acute lung injury and ARDS were identified. Both meta-analyses reported a risk of renal impairment associated with iNO treatment in adults.<sup>8,29</sup> As the analysis of renal impairment was based only on four of all randomized, controlled trials, although the four studies enrolled 72% of the entire study collective, this result has to be interpreted with caution.<sup>8</sup>

A conclusive explanation for this observation cannot be offered. Inhibition of mitochondrial and enzymatic function, membrane damage or alterations of deoxyribonucleic acid have been discussed.<sup>8</sup> Renal failure in association with iNO treatment has not yet received much attention and needs further investigation.

## Safe Inhaled Nitric Oxide Therapy and Contraindications

Safe iNO therapy requires continuous analysis of NO and NO<sub>2</sub> concentrations, closely integrated respiratory and hemodynamic monitoring of the patient, periodic analyses of methemoglobin levels, use of certified NO delivery systems and tanks, and administration of the lowest concentration required. Rebound problems after withdrawal of iNO demand skill and endurance during weaning of the drug.



Contraindications to iNO therapy are based on its toxicology. In clinical practice, methemoglobin formation and elevated NO<sub>2</sub> levels are of interest, although both are uncommon during iNO therapy with normal therapeutic doses. These complications may become clinically important when patients are receiving higher doses of iNO. In patients with high levels of methemoglobin, additional iNO therapy may not be indicated and in patients with methemoglobin reductase deficiency, iNO therapy is contraindicated. The clinical risks of bleeding problems may be neglected. Data from large-scale meta-analyses in adults, children and infants did not reveal an elevated risk of coagulopathy associated with iNO. In patients with a hemorrhagic diathesis or coagulation or bleeding problems, however, iNO therapy should be well considered. There are indications that iNO might compromise renal function in adults with acute lung injury or ARDS. To date, this issue remains open.

## The Future

What are upcoming scientific challenges in studying this extraordinary substance? What about the development of alternative NO therapies for the respiratory system?

First, more high-quality studies of diseases in which iNO therapy is of benefit are needed. These studies should evaluate not only survival rates with iNO therapy, but also outcome variables such as quality of life, cognitive impairment, and lung function. Dose–response studies for the different indications of iNO therapy are needed, and safety of long-term application of iNO should be investigated. The responder–nonresponder problem also needs to be probed.

Basic scientific work is needed to investigate the problem of the physiologic autoinhalation of endogenous NO from the paranasal sinuses, which is excluded in intubated and ventilated patients. It is assumed that some concentration of autoinhaled NO in the airways is necessary and protective against infection. During intubation and mechanical ventilation, low doses of iNO may restore the physiologic airway concentration of NO and protect against ventilator-associated pneumonia. It is possible that a patient's own nasal NO may be redirected into the inspiratory limb of the ventilatory circuit.

Therapy with iNO is expensive and technically demanding. Future research may yield cheaper, alternative drugs that selectively supply NO in ventilated lung regions. Inhalation of prostacyclins, which also produces selective pulmonary vasodilation, may be a cheaper alternative, although further studies are necessary to advocate inhaled prostaglandins as a promising therapeutic option.

Is sildenafil the new iNO? In neonates with PPHN, sildenafil, administered orally or intravenously yielded promising results.<sup>49,50</sup> In most cases oxygenation improved significantly, and a randomized controlled trial of fifty-one term infants with PPHN reported a decrease in mortality.<sup>50</sup> It's a moot question, however, whether sildenafil is of benefit in adult patients with hypoxic respiratory failure. Limited data, however, suggest that sildenafil, does decrease PAP, but increases shunt fraction, which may make the drug unsuitable for treatment of patients with ARDS.<sup>51</sup>

The suspense continues. Will iNO survive in the long run or will it be replaced by intelligent, selective intravenous vasodilators?

## Summary and Conclusion

NO is an important signaling molecule in the respiratory system. It modulates perfusion of lung units and it can prevent inflammatory processes. Administered by inhalation, low doses of NO selectively dilate pulmonary blood vessels in ventilated lung areas and redistribute blood flow from atelectatic to aerated lung units. When lower concentrations of iNO are applied, almost no systemic vasodilation can be observed because most of the NO rapidly binds to hemoglobin and is thus inactivated. The overall effect of NO inhalation is a sustained reduction of elevated PAP and improvement in oxygenation if a ventilation–perfusion mismatch of the lung is present. iNO is used to treat several cardiopulmonary disorders that are associated with hypoxia or pulmonary hypertension; its main application, however, is in neonatology. In infants with PPHN, iNO significantly reduces the need for ECMO. In no instance, however, was NO clearly shown to enhance long-term survival. There is no end in sight for the future role of NO inhalation. What we urgently need are more high-quality studies of the diseases in which iNO therapy is effective, and, hopefully, to demonstrate higher survival rates or at least improved secondary outcome variables, for example, reduced need of extrapulmonary oxygenation, fewer neurodevelopmental sequelae, or improved quality of life.

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